

MD. HASAN IMON

Jr. AI/ML Engineer | Jr. AI/ML Developer

Address: Savar, Dhaka, Bangladesh **Email:** emon.mlengineer@gmail.com **Portfolio:** md-emon-hasan.github.io/My-Resume
Linkedin: linkedin.com/in/md-emon-hasan-695483237/ **GitHub:** github.com/Md-Emon-Hasan **WhatsApp:** +8801834363533

SUMMARY

AI/ML Engineer specializing in Generative AI, NLP, and Machine Learning, with expertise in Deep Learning and developing production-grade multi-agent systems, RAG pipelines, and tool-augmented AI architectures. Skilled in fine-tuning LLMs and applying NLP techniques such as transformers and language modeling to build robust ML/DL solutions. Experienced in FastAPI, Docker, and CI/CD for designing and deploying scalable, end-to-end AI systems that enhance decision-making, drive automation, and deliver measurable business outcomes.

EXPERIENCE

Jr. Machine Learning Engineer - Hi-Tech Parks Ltd. | Dhaka, Bangladesh | June 2025 – Sept 2025

- Performed data preprocessing, feature engineering, and performance optimization for robust pipelines.
- Developed RAG pipelines and multi-agent AI systems applications, ensuring modularity and scalability.
- Collaborated with AI & MLOps teams to integrate models into CI/CD with Docker and FastAPI/Flask.

Research Assistant (AI/ML) - City University Bangladesh | May 2025 – Sept 2025

- Authored a research paper "Boosting Security: Gradient Boosting Models for Advanced Malware Detection".
- Designed full pipeline and state-of-the-art performance with leveraging ensemble & boosting techniques, achieved 80% accuracy, F1-score 0.80 which outperforming classical ML models.

PROJECTS

MediGenius: Multi-Agent Medical Assistant – [GitHub](#)

Build an AI-driven medical assistant with Multi agent communication control.

- Achieving 82% clinical alignment, reducing misinformation by ~60% and improving patient query resolution rate by ~75%, driving higher trust, Context-aware, multi-turn consultations and adoption with enabling real-time empathetic medical support AI-driven healthcare consultation.
- Architected a modular, tool-augmented agentic ecosystem integrating Planner, Retriever, Answer Generator, and Fallback Agents with medical RAG over verified PDFs, SQLite with long term memory, and a Flask, HTML, CSS and JS deployed via Dockerized CI/CD pipeline.

InformaTruth: Fine-tuned AI-Driven News Authenticity Analyzer – [GitHub](#)

Build explainable fake news detection system using Fine-tuned datasets.

- Multi-agent fact-checking system that automatically verifies news, PDFs, and web content, achieving ~70% accuracy and F1 ~69% on the LIAR dataset. Implemented with RoBERTa fine-tuning, LangGraph, RAG pipelines, fallback retrieval agents, FastAPI endpoints, Dockerized for deployment with Web UI.
- Boosted business impact and reliability by integrating memory, multi-agent tool routing, and fallback workflows, improving query coverage to 95% and answer reliability by ~60%, enabling automated fact verification at scale.

AutoDocThinker: Intelligent Search Engine with Agentic RAG System – [GitHub](#)

Build an intelligent document QA system using Agentic RAG

- Developed Q&A system that provides accurate answers from PDFs, DOCX, TXT, and URLs, achieving ~95% context retrieval coverage and reducing fallback calls to Wikipedia to ~5%, enabling faster and more reliable information access for enterprise users.
- Stateful multi-agent workflow with dynamic tool routing, supporting ingestion, retrieval, LLM answering, and fallback agents. Engineered with vector-based similarity search, recursive text splitting improving query processing efficiency by ~40% while handling 50+ simultaneous documents with Web apps.

TrueWealth AI: Your AI-Powered Financial Strategist – [GitHub](#)

Multi-agent financial advisor with Query breakdown & multi-stage planning.

- Built an AI-powered multi-agent financial advisor that delivers reliable, real-time investment insights by combining LLM reasoning, Agentic RAG and fallback web/news search. Achieving 99% query coverage with <3s P95 latency, reducing manual research effort for clients.
- Enhanced with LangGraph orchestration, HuggingFace embeddings, ChromaDB, Groq LLM, and modular agents (Planner, Retriever, Executor, Memory, Tool Router), deployed via Docker + FastAPI + Web UI, ensuring scalability, resilience, and enterprise readiness.

SKILLS

- ✓ **Programming & Data:** Python, Data Analytics, Pandas, NumPy, Matplotlib, Seaborn.
- ✓ **MLOps & Backend:** REST APIs, FastAPI, Flask, Git/GitHub, OOP, Docker, Postman
- ✓ **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, PyTorch, TensorFlow, Keras, Scikit-learn
- ✓ **NLP & Large Language Models (LLMs):** Text Classification, Embeddings, Hugging Face, NLTK, spaCy, HuggingFace Transformers, GPT, LLaMA, Mistral, Claude, Custom Fine-tuning (PEFT, LoRA, QLoRA)
- ✓ **AI Agents & Multi-Agent Systems:** LangChain, LangGraph, AgentOps, CrewAI, LlamaIndex, Agentic RAG Pipelines, Agent Orchestration, Tool-Augmented, Memory-Augmented, Prompt Engineering, HITL
- ✓ **Vector Databases & Storage:** FAISS, ChromaDB, Pinecone, Weaviate, MySQL, MongoDB
- ✓ **Cloud & Deployment:** Docker, CI/CD pipelines, Render

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science & Engineering

City University, Bangladesh (2022 - 2025)

CERTIFICATION

- **Machine Learning Specialization** by Stanford University (Coursera)
- **Programming for Everybody** by University of Michigan (Coursera)
- **Google UX Design Specialization** by Google (Coursera)
- **Google IT Support Specialization** by Google (Coursera)
- **IBM Cybersecurity Analyst Specialization** by IBM (Coursera)